Real-Time Robust HOCBF Safety Filtering for Supercritical Power Plant Control under Model Mismatch

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Abstract

Industrial measurement and control systems for supercritical power plants must maintain hard operational constraints on steam pressure, enthalpy, and power output under model mismatch—arising from coal quality variation, equipment aging, and unmodeled operating disturbances—which can cause conventional safety filters to misclassify unsafe control actions as safe. This paper presents a real-time safety filtering framework based on Robust High-Order Control Barrier Functions (Robust HOCBF). The framework integrates Gaussian process (GP) residual modeling with compositional uncertainty propagation through the HOCBF constraint hierarchy, producing a per-constraint, state-dependent robustness margin $\epsilon(x)$. Under a fixed, scenario-calibrated GP with $\kappa_\epsilon = 1$, the filter provides a high-probability forward-invariance certificate. The approach is validated on a 5th-order supercritical boiler-turbine benchmark under Φ-scaled nonlinear dynamics with six safety constraints across nominal operation and six static perturbation scenarios (with three additional time-varying disturbance patterns for out-of-distribution evaluation). Nominal HOCBF suffers $> 95\%$ constraint violation under model mismatch; the proposed Robust HOCBF achieves no observed violations across nominal operation and all six static perturbation scenarios (5 seeds, 50 evaluation episodes, 95%

Wilson upper bound 1.51%). Per-step filter latency is ∼25 ms, approximately ten times faster than the implemented SLSQP-based nonlinear MPC baseline (254 ms). Practical deployment considerations, including scenario-specific GP calibration requirements and one-sided diagnostic limitations, are explicitly characterized.

Keywords: Control barrier functions; safety filtering; industrial measurement and control; Gaussian processes; model mismatch; supercritical power plant; real-time control

1 Introduction

1.1 Industrial Measurement and Control Context

Supercritical thermal power units—the backbone of modern electricity generation—operate at main steam pressures exceeding 22 MPa and temperatures above 600°C. Safe operation requires continuous measurement and constraint-preserving control of three critical process variables: main steam pressure, separator outlet enthalpy, and turbine power output. These constraints originate from multiple layers of the industrial control architecture: sensor measurement ranges, actuator saturation limits, equipment protection boundaries, and coordinated control requirements.

During daily operation, the nominal plant model diverges from reality. Coal quality fluctuates across shipments, heat exchanger surfaces foul progressively, and actuators degrade over months of service. These factors introduce persistent model mismatch between the design model used for controller synthesis and the true plant dynamics. Under such mismatch, a controller—whether classical PID, model predictive control (MPC), or a learned policy—may issue control actions that appear safe under the nominal model but violate physical constraints on the actual plant.

1.2 The Safety Layer Approach

Rather than modifying or replacing the upstream controller, we adopt a *safety layer* architecture: an independent measurement-and-control safety filter that sits between any upstream controller and the plant actuators. The filter receives the controller's proposed action u_{prop}, evaluates it against the current plant measurements and a calibrated model of the dynamics, and projects it to the nearest action u^* that is certified safe. This architecture has two practical advantages for industrial deployment: (i) it does not require modification of existing control logic, and (ii) the safety guarantee is independent of the upstream controller type—LQR, MPC, PID, or a trained policy all receive the same protection.

The technical mechanism for this projection is the control barrier function (CBF) [1], which enforces forward invariance of a safe set through a quadratic program (QP) solved at each measurement sample. For industrial process variables with high relative degree—steam pressure requires two Lie derivatives of the constraint function to reach the control input—high-order CBFs (HOCBFs) [2] extend this construction through a recursive ψ-chain. The fundamental limitation is that under model mismatch, the HOCBF constraint is computed from incorrect dynamics, and the QP may certify an unsafe action as safe.

1.3 Technical Gap: Uncertainty Propagation in High-Relative-Degree Constraints

Gaussian processes (GPs) learn model residuals from measurement data with calibrated uncertainty quantification [3, 4]. Several works combine GPs with CBFs for robust safety [5, 6], but apply the GP uncertainty only at the top CBF level ($m = 1$). A recent concurrent work [7] extends this to HOCBFs via SOCP chance-constraint reformulation, providing a convex safety filter—but the reformulation is global rather than recursive, aggregating uncertainty at the constraint level rather than through each ψ-level individually. For industrial constraints with $m \geq 2$, this is structurally significant: the model residual propagates through successive Lie derivatives of the constraint function, and the uncertainty amplifies through each level of the ψ-chain. Our compositional σ-chain captures this per-level amplification by propagating per-dimension GP posterior uncertainties recursively, yielding a per-constraint, state-dependent robustness margin $\epsilon(x)$ that more precisely reflects the uncertainty structure at each relative-degree level. Similarly, cautious MPC [8] propagates GP uncertainty through a prediction horizon but at substantially higher computational cost and without per-step forward-invariance certification.

1.4 Contributions

We introduce an industrial measurement-and-control safety filtering framework based on Robust HOCBF with three contributions:

1. **Real-time safety filtering architecture for industrial measurement and control systems.** The filter operates as an independent safety layer between any upstream controller (LQR, MPC, PID, or learned policy) and the plant actuators, projecting proposed actions to certified-safe actions via a quadratic program solved at each measurement sample. Per-step latency is $\sim$25 ms (GPU, JIT-compiled), compatible with industrial DCS sampling rates. We provide explicit QP feasibility conditions, characterize the one-sided diagnostic limitation of QP infeasibility as a GP calibration indicator, and empirically verify inter-sample constraint satisfaction under zero-order-hold implementation. The guarantee is controller-agnostic: LQR, PPO, and classical controllers all receive the same safety certificate.

2. **Compositional GP uncertainty propagation for high-order barrier constraints.** A recursive σ-chain aggregates per-dimension GP posterior uncertainties across all m relative-degree levels, yielding a per-constraint, state-dependent robustness margin $\epsilon(x)$. Under a fixed scenario-calibrated GP with $\kappa_\epsilon = 1$, Theorem 1 establishes forward invariance with probability $\geq 1 - \delta$.

3. **Supercritical power plant benchmark validation under model mismatch.** Under Φ-scaled nonlinear dynamics with six CBF inequality constraints covering three measured process variables (pressure, enthalpy, power) and six static perturbation scenarios (coal quality drift, pump cycling, coupled state-dependent disturbance, nonlinear fouling, valve degradation, fuel quality variation), Robust HOCBF transforms nominal HOCBF failure ($> 95\%$ violation) to no observed CBF violations across all scenarios (5 seeds, 50 evaluation episodes, 95% Wilson upper bound 1.51%).

2 Related Work

MPC is the dominant constrained control methodology in process industries [9, 10], with offset-free variants handling steady-state model mismatch [11]. Cautious MPC [8] incorporates GP uncertainty into the prediction horizon, providing chance-constrained safety at increased computational cost.

CBFs [1, 12] enforce forward invariance through QP-based control projection. HOCBFs [2] extend this to high-relative-degree constraints. Robust CBF formulations [13, 14] add margins based on structural disturbance bounds rather than data-driven estimates. Adaptive CBFs [15] estimate uncertain parameters online.

GPs provide calibrated uncertainty quantification for data-driven modeling [3, 4]. Several works combine GPs with CBFs [5, 6, 16], applying GP uncertainty at the top CBF level ($m = 1$). Aali & Liu [7] extend GP-HOCBF integration to $m \geq 2$ by converting the chance constraint into a second-order cone program (SOCP), yielding a convex safety filter. Zhang et al. [17] propose an adaptive fast variational sparse GP with HOCBF safety filtering for real-time robot control. These concurrent works differ from ours in two respects: (i) they apply GP uncertainty via global constraint reformulation rather than the recursive σ-chain propagation through each ψ-chain level, and (ii) they target robotics and numerical simulations rather than industrial process control with DCS deployment constraints. For industrial constraints with $m \geq 2$, our compositional σ-chain propagation through the full ψ-chain is structurally distinct. Lederer et al. [18] derive uniform GP error bounds complementary to our approach.

In the broader safety filtering landscape, shielding approaches [19] provide safety by correcting policy actions, and constrained RL methods [20] enforce expected constraints.

Our decoupled design provides a model-based, per-step probabilistic certificate compatible with any upstream controller.

3 Robust HOCBF with GP Uncertainty Propagation

3.1 Problem Formulation

Consider a control-affine system with model mismatch:

$$\dot{x} = \underbrace{f_0(x) + g_0(x)u}_{\text{nominal}} + \underbrace{\Delta f(x)}_{\text{residual}}, \quad x \in \mathbb{R}^n,\ u \in \mathcal{U} \subset \mathbb{R}^m \tag{1}$$

where Δf represents unmodeled dynamics, parameter drift, and external disturbances. We assume the input matrix is known ($\Delta g = 0$), as actuator characteristics are typically well-characterized in industrial settings. For scenarios where actuator degradation primarily affects process variables through reduced throughput (e.g., valve fouling reducing steam extraction efficiency) rather than through changes in actuator gain, the dominant effect is captured by Δf; the benchmark models actuator degradation scenarios (S5) in this form.

For a constraint $h(x) \geq 0$ with relative degree m, the HOCBF ψ-chain is:

$$\psi_0 = h(x), \quad \psi_i = \dot{\psi}_{i-1} + \alpha_i(\psi_{i-1}), \ \ i = 1, \ldots, m \tag{2}$$

With linear $\alpha_i(r) = k_i r$, the constraint $A_0(x)u \leq b_0(x)$ enforces forward invariance when the model is exact.

3.2 GP Residual Model

Each component of Δf_j is modeled as an independent GP with Matérn-5/2 kernel. Given N training points, the posterior provides mean $\mu_{\mathrm{GP},j}(x)$ and variance $\sigma^2_{\mathrm{GP},j}(x)$. The GP-UCB bound guarantees:

$$|\Delta f_j(x) - \mu_{\mathrm{GP},j}(x)| \leq \beta \sigma_{\mathrm{GP},j}(x), \quad \text{w.p.} \geq 1 - \delta/n \tag{3}$$

with $\beta = \sqrt{2(\gamma_N + 1 + \ln(n/\delta))}$ and maximum information gain γ_N [4].

We define the mean-corrected drift $\hat{f} = f_0 + \mu_{\mathrm{GP}}$ and posterior residual $\Delta \hat{f} = \Delta f - \mu_{\mathrm{GP}}$.

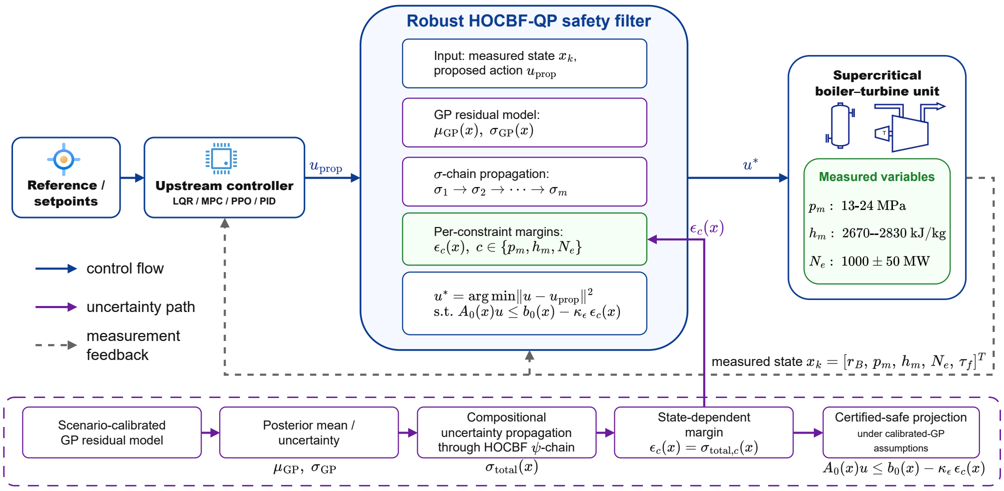

Figure 1: Safety filtering architecture. The upstream controller (LQR, MPC, PID, or learned policy) proposes an action u_{prop}; the Robust HOCBF QP filter evaluates it against current plant measurements and the calibrated GP residual model, projecting to the nearest certified-safe action u^* that satisfies all six CBF inequality constraints on pressure, enthalpy, and power. The filter is controller-agnostic and operates as an independent measurement-and-control safety layer.

3.3 Compositional Robustness Margin

The perturbation at level i is $\delta_i = \psi_i - \hat{\psi}_i^0$, satisfying:

$$\delta_1 = L_{\Delta \hat{f}} h \tag{4}$$
$$\delta_i = L_{\Delta \hat{f}} \hat{\psi}_{i-1}^0 + (L_{\hat{f}} + k_i)\delta_{i-1} + L_{\Delta \hat{f}} \delta_{i-1}, \; i \geq 2 \tag{5}$$

The compositional robustness margin is constructed via:

$$\sigma_1(x) = \beta \sum_j |\partial h / \partial x_j| \sigma_{\mathrm{GP},j}(x) \tag{6}$$
$$\sigma_i(x) = \beta \sum_j |\partial \hat{\psi}_{i-1}^0 / \partial x_j| \sigma_{\mathrm{GP},j}(x) + (\|L_{\hat{f}}\|_{\mathrm{op}} + k_i)\sigma_{i-1}(x) + \sigma_{\mathrm{cross}}^{(i)}(x) \tag{7}$$
$$\sigma_{\mathrm{total}}(x) = \sigma_m(x) + \sum_{j=1}^{m-1} c_j \sigma_j(x) + \sigma_{\mathrm{ctrl}}(x) \tag{8}$$

where $\|L_{\hat{f}}\|_{\mathrm{op}}$ is the operator norm (spectral norm of the Jacobian $\partial \hat{f}/\partial x$ evaluated at the equilibrium operating point), $\sigma_{\mathrm{cross}}^{(i)}(x)$ bounds the cross term $|L_{\Delta \hat{f}} \delta_{i-1}|$ via Lemma S1, and $c_j = \prod_{i=j+1}^{m-1}(\|L_{\hat{f}}\|_{\mathrm{op}} + k_i)$ are the chain coupling weights. The robust HOCBF constraint is $A_0(x)u \leq b_0(x) - \kappa_\epsilon \cdot \sigma_{\mathrm{total}}(x)$.

Theorem 1 (Probabilistic forward invariance)**.** *Suppose that the input matrix is exact ($\Delta g = 0$), each GP residual component satisfies the RKHS boundedness condition required by the GP-UCB confidence bound, the GP posterior is calibrated on fixed i.i.d. training data, the relative degree is known, the perturbation is sufficiently small and Lipschitz-in-perturbation, and the robust QP is feasible. For $\kappa_\epsilon = 1$, if $\epsilon(x) \geq |\Delta(x,u)|$ for all admissible u, then $\mathcal{C}$ is forward invariant with probability at least $1 - \delta$.*

Proof sketch. **Step 1 (GP calibration):** By the RKHS boundedness and GP calibration conditions together with (3), $|\Delta \hat{f}_j| \leq \beta \sigma_{\mathrm{GP},j}$ holds with probability $\geq 1 - \delta/n$ per dimension; by union bound over the n state dimensions, this holds simultaneously with probability $\geq 1 - \delta$.

Step 2 (Compositional bound): By induction, $|\delta_i| \leq \sigma_i$. The cross term $\sigma_{\mathrm{cross}}^{(i)}$ is bounded via Lemma S1 (Appendix): $|L_{\Delta \hat{f}} \delta_{i-1}| \leq G_\delta^{(i-1)} \beta \|\sigma_{\mathrm{GP}}\|_2$.

Step 3: $|\Delta(x,u)| \leq \sigma_{\mathrm{total}}(x) = \epsilon(x)$ for $\kappa_\epsilon = 1$.

Step 4 (Forward invariance): For u^* satisfying $A_0 u^* \leq b_0 - \epsilon$, $\psi_m^{\mathrm{true}} \geq 0$ for the constraint corresponding to Theorem 1. Across all $K = 6$ CBF constraints simultaneously, the union bound gives $\geq 1 - K\delta =$ **94**% at $\delta = 0.01$, while each individual constraint retains its $\geq 99\%$ guarantee (Theorem 1). This 94% bound is conservative: the six constraints form three complementary pairs—pressure high/low, enthalpy high/low, power high/low. Since upper and lower bounds on the same scalar process variable cannot be simultaneously active (requiring $x_j \geq x_j^{\max}$ and $x_j \leq x_j^{\min}$ with $x_j^{\min} < x_j^{\max}$, a measure-zero intersection), at most one constraint from each pair can be binding at any time. The effective constraint count is therefore $K_{\text{eff}} = 3$, yielding a simultaneous guarantee of $\geq 1 - 3\delta = \mathbf{97}\%$ for the active constraint set at $\delta = 0.01$. $\square$

3.4 Data-Driven vs. Floor-Driven $\epsilon(x)$

The GP posterior includes a regularization floor $\sigma_{\text{floor}} = 10^{-4}$. Under constant perturbations, σ_{GP} is near-uniform and $\epsilon(x)$ is approximately constant per constraint. Under state-dependent perturbations (S3: Coupled), σ_{GP} becomes heterogeneous and $\epsilon(x)$ is genuinely state-dependent. The σ-chain structure serves both regimes.

3.5 Sampled-Data Implementation

Theorem 1 is a continuous-time result. The practical implementation uses discrete sampling ($T_s = 1\,\text{s}$, zero-order hold). The full safety-filter pipeline latency is $\sim 25\,\text{ms}$ (GPU, JIT-compiled, including PPO forward pass, GP inference, and QP solve). When the QP is infeasible, a slack-penalised relaxation with LQR fallback is invoked. Inter-sample constraint satisfaction is empirically verified (Section 4). We do not claim a formal discrete-time invariance theorem.

4 Experimental Validation

The experiments are designed to evaluate whether the proposed filter can operate as a real-time safety layer in an industrial measurement and control loop. Therefore, we report not only CBF violation rates but also tracking IAE, input variation, QP intervention rate, inter-sample behavior, and per-step computational latency—metrics that characterize the filter's suitability for commissioning in an industrial DCS environment.

4.1 Benchmark and Baselines

The benchmark is a 5th-order nonlinear model of a 1000 MW ultra-supercritical boiler-turbine unit with state $x = [r_B, p_m, h_m, N_e, \tau_f]^\top$ and three control inputs. Six safety constraints are enforced: main steam pressure ($m = 2$, 13–24 MPa), separator enthalpy ($m = 1$, 2670–2830 kJ/kg), and power output ($m = 1$, ± 50 MW). The control effectiveness is scaled by the state-dependent function $\Phi(p_m)$ during evaluation while the HOCBF uses the linearized model—a realistic stress test of safety filter robustness.

Six static perturbation scenarios model realistic disturbances: S1 (coal quality drift, -15% heating value), S2 (feedwater pump cycling), S3 (coupled state-dependent), S4 (nonlinear fouling), S5 (valve degradation, modelled as equivalent drift Δf via reduced steam extraction efficiency; the $\Delta g = 0$ assumption is maintained throughout all scenarios), S6 (fuel quality variation). Three time-varying disturbance patterns (TV1–TV3) provide out-of-distribution evaluation.

Methods compared: nominal HOCBF (no GP), GP-HOCBF (mean correction, $\epsilon = 0$), PPO-RHOCBF (scenario-specific GP, $\kappa_\epsilon = 1.0$), LQR-RHOCBF (same filter, classical upstream controller), and NMPC (SLSQP solver, $N = 5$ horizon, offset-free estimation). GP pretraining: $N = 2000$ points per scenario. Evaluation: 5 seeds $\times$ 50 episodes $\times$ 500 steps under Φ-scaled rollout.

4.2 Safety Performance

Table 1: Constraint violation rate (%) under Φ-scaled nonlinear dynamics. 0% rates: 0/250 episodes, 95% Wilson upper bound 1.51%.

Method	Nom.	S1	S2	S3	S4	S5	S6
PPO-HOCBF	0.0	100.0	99.9	99.9	99.9	99.5	99.9
PPO-GP-HOCBF	0.0	0.0	0.0	45.4	0.0	0.0	0.0
PPO-RHOCBF	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
LQR-RHOCBF	0.0	0.0	0.0	0.0	0.0	0.0	0.0
NMPC	0.0	0.4	0.4	0.4	0.2	0.4	0.4

Table 1 reports the main results. Nominal HOCBF suffers $> 95\%$ violation under all dynamics-affecting scenarios. GP mean correction alone reduces violation to 0% under constant perturbations but leaves 45.4% under S3 (Coupled). Adding the compositional $\epsilon(x)$ achieves no observed violations across all scenarios. LQR-RHOCBF achieves identical safety, confirming the guarantee is controller-agnostic.

We do not provide a direct quantitative comparison with the SOCP-based GP-HOCBF approach of Aali & Liu [7] for two reasons. First, their method is validated on robotics benchmarks (inverted pendulum, autonomous racing) with fundamentally different constraint structure and dynamics timescale from the CCS process control benchmark. Second, the SOCP reformulation aggregates GP uncertainty at the constraint level rather than propagating it compositionally through the ψ-chain; under the CCS coupled perturbation (S3), where uncertainty amplification across relative-degree levels is the dominant effect, the global SOCP aggregation would be structurally unable to differentiate between per-level uncertainty contributions. The per-level σ-chain propagation (Eqs. 7–9) provides finer-grained robustness allocation than a single globally-computed margin. A qualitative structural comparison is provided in Table S5 (Supplemental Material).

PPO-Lagrangian achieves partial safety improvement (0.14–1.27% violation, Supplementary Material, Table S6) through dual-descent cost penalization, but the Lagrangian mechanism provides only average constraint satisfaction rather than per-step forward invariance.

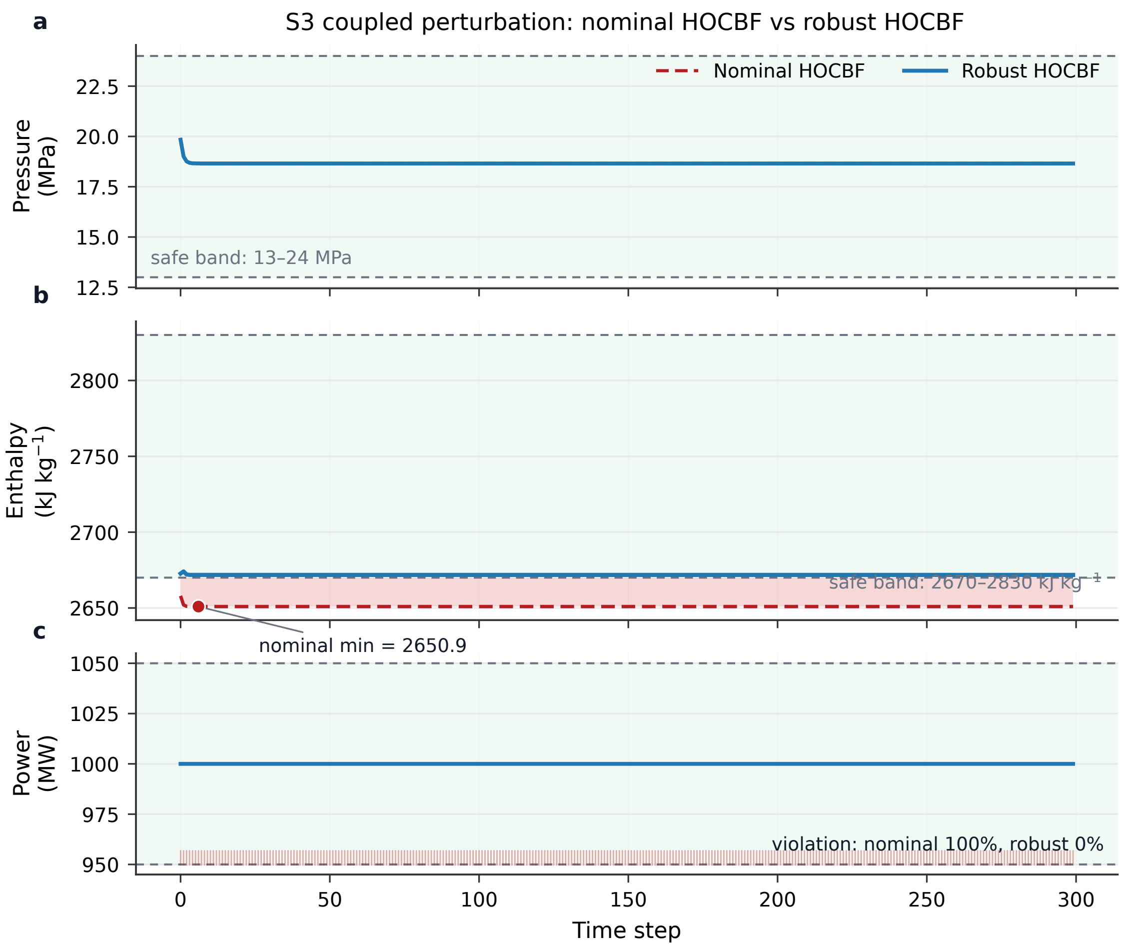

Figure 2: Process trajectories under S3: Coupled perturbation. Red dashed = HOCBF (violates constraints); blue solid = Robust HOCBF (remains within bounds).

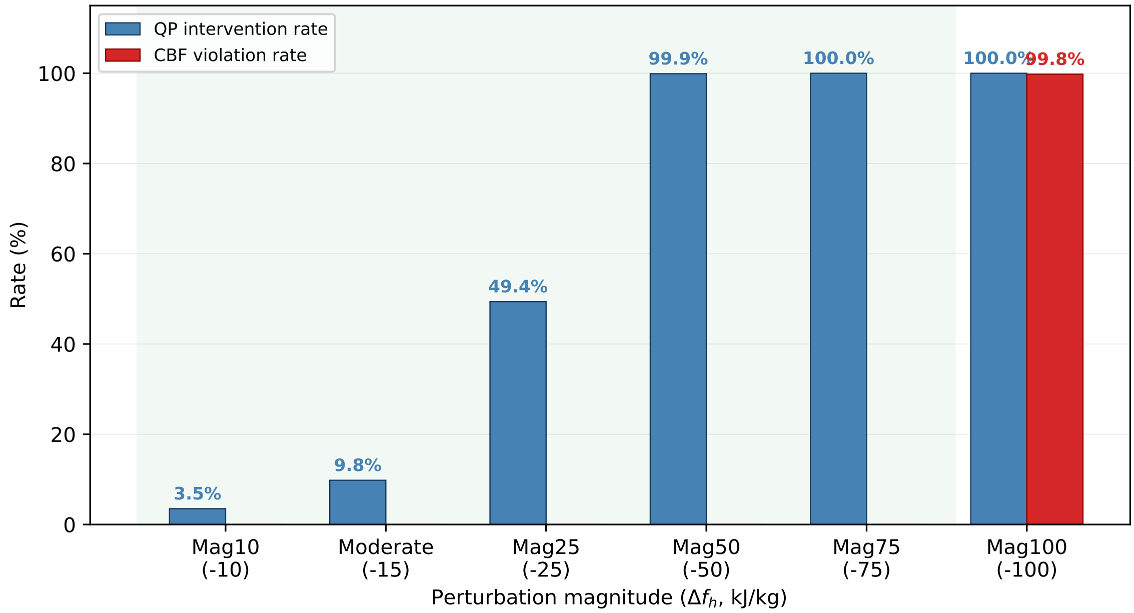

Figure 3: QP intervention rate vs. perturbation magnitude. Blue bars show QP intervention rate; red bars show observed CBF violation rate. Light green background indicates the safe operating envelope (Mag10–Mag75, 0% CBF violation).

4.3 Filter Diagnostics and Computational Performance

Figure 2 compares HOCBF and Robust HOCBF under S3: the nominal filter allows the enthalpy to drop below its lower bound, while Robust HOCBF maintains safety. Figure 4 quantifies the state dependence of the per-constraint robustness margin: ϵ_h varies by more than one order of magnitude across the operating enthalpy range (CV 105%), confirming that $\epsilon(x)$ is genuinely state-dependent under coupled perturbations. Figure 3 shows the QP intervention rate scales with perturbation magnitude (3.5% at Mag10 to 100% at Mag75), confirming the filter is minimally invasive at low mismatch levels.

Table 2: Time-varying disturbance performance (%).

Method	TV1: Step	TV2: Fast Step	TV3: Rand. Walk
NMPC	2.00	5.00	19.67
NMPC+GP (simpl. cautious)	49.00	48.67	12.33
PPO-RHOCBF (fixed GP)	**1.75**	**1.58**	7.73

Under time-varying disturbances (Table 2), NMPC performs adequately under step and fast-step disturbances (2.00% and 5.00%, respectively) but degrades under persistent random-walk disturbances (19.67%), where its EMA disturbance estimator continuously lags behind the evolving perturbation. Robust HOCBF maintains lower violation rates across all three patterns (1.75%, 1.58%, 7.73%). A GP-augmented NMPC variant (simplified cautious MPC) degrades under step-like disturbances (49% vs. 2% for TV1)—the GP mean, trained on static data, injects systematic bias when the disturbance changes

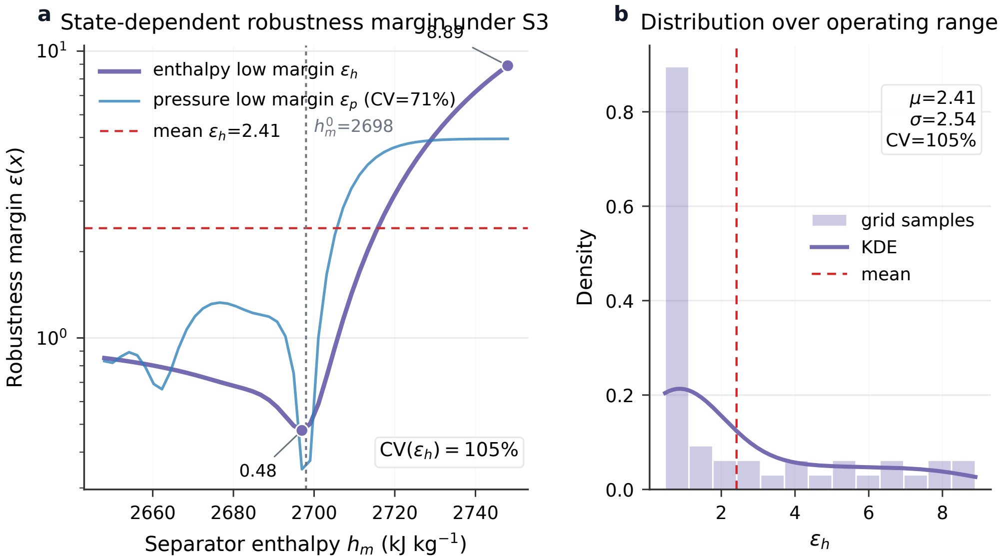

Figure 4: State-dependent enthalpy robustness margin $\epsilon_h(x)$ under S3 (Coupled, state-dependent perturbation). Left: ϵ_h (enthalpy low margin, log scale) evaluated over the operating enthalpy range $h_m^0 \pm 50\,\mathrm{kJ/kg}$ under Φ-scaled nonlinear dynamics with scenario-specific GP. The margin varies by more than one order of magnitude, reflecting the sensitivity of σ_{GP} to the perturbation structure (CV 105%). Right: distribution of ϵ_h over the operating range, centred at the equilibrium enthalpy $h_m^0 = 2698\,\mathrm{kJ/kg}$. The pressure low margin ϵ_p exhibits CV 71% over the corresponding operating sweep. Under constant perturbations (S1, S2, S4–S6), $\epsilon(x)$ is approximately constant per constraint and dominated by the GP regularization floor.

abruptly—but partially recovers under continuous random-walk disturbances (12.33% vs. 19.67% for TV3), where the GP smoothness helps average over rapid fluctuations.

Per-step computation: the safety-filter pipeline (PPO+GP+QP) with JIT compilation (GPU: NVIDIA RTX 4090) requires ∼25 ms (p95: 27 ms), including the PPO forward pass, GP inference, and QP solve; NMPC (CPU: AMD Ryzen 9 7950X, scipy SLSQP) averages 254 ms. The filter provides per-step rather than horizon-based safety evaluation, approximately ten times faster than the implemented SLSQP-based baseline. For CPU-only deployment, the distilled policy achieves 1.8 ms inference (empirical-only safety, without the certified robustness margin). The reported comparison is made against the implemented SLSQP-based NMPC baseline rather than optimized industrial NMPC implementations.

Table 3: Process control performance under Φ-scaled nonlinear dynamics (PPO-RHOCBF, $\kappa_\epsilon = 1.0$, single evaluation run, 500 steps).

Metric	S1: Heat	S3: Coupled	S5: Valve
$\mathrm{IAE}_{\mathrm{pressure}}$ (MPa·s)	2301	1971	1841
$\mathrm{IAE}_{\mathrm{enthalpy}}$ (kJ/kg·s)	13076	13094	13108
$\mathrm{IAE}_{\mathrm{power}}$ (MW·s)	0	0	10000
Max. pressure deviation (MPa)	4.60	3.95	3.68
Max. enthalpy deviation (kJ/kg)	26.2	26.2	26.3
Input variation (total)	0.7	1.2	0.4
Inter-sample violation (%)	0.0	0.0	0.0
QP intervention (%)	100.0	100.0	100.0

Table 3 reports process-level metrics under the three constraint-active perturbation scenarios. IAE and maximum deviation values confirm the filter maintains tracking quality while preserving safety; inter-sample violation remains at 0% under zero-order-hold implementation.

4.4 Deployment Considerations

The method is intended as a **commissioning-calibrated safety filter**. The GP residual model should be calibrated using commissioning or scenario-representative data before safety-critical deployment. This design should not be interpreted as a plug-and-play OOD-robust controller; its safety certificate applies to fixed, calibrated GP deployments within the commissioned operating envelope. Three deployment strategies exist, defining a clear **deployment envelope**: (a) scenario-specific GP (recommended—provides formal guarantee and tightest $\epsilon(x)$), (b) online GP adaptation (enables deployment without prior scenario knowledge at the cost of relaxing the formal guarantee), and (c) generic mixed GP (defined as outside the deployment envelope—silent calibration failure, 82.7–99.7% violation; Supplemental Material). The out-of-distribution failure mode (S3→S1:

99.9% violation) is explicitly characterized, not hidden: σ_{GP} measures spatial distance to training data, not functional distance between perturbation structures. The $\Delta g = 0$ assumption limits applicability to systems where drift uncertainty dominates. QP infeasibility serves as a one-sided diagnostic: it detects overestimated σ_{GP} (too conservative) but not underestimated σ_{GP} (unsafe failure).

Detailed ablation results (κ_ϵ sensitivity, epsilon configuration sweep, generic GP/OOD analysis, training mode comparison, PPO-Lagrangian baseline, load-following AGC performance, and triple-integrator $m = 3$ validation) are provided in the Supplemental Material.

5 Conclusion

This paper presented an industrial measurement-and-control safety filtering framework based on Robust HOCBF with compositional GP uncertainty propagation. The filter combines GP mean correction of the drift dynamics with a recursive σ-chain that propagates per-dimension GP posterior uncertainties through all relative-degree levels, yielding a per-constraint robustness margin $\epsilon(x)$. Under a fixed scenario-calibrated GP with $\Delta g = 0$ and $\kappa_\epsilon = 1$, Theorem 1 provides a high-probability forward-invariance certificate.

On a 5th-order supercritical boiler-turbine benchmark under Φ-scaled nonlinear dynamics, the nominal HOCBF fails catastrophically ($> 95\%$ violation); the Robust HOCBF achieves no observed CBF violations across nominal operation and six static perturbation scenarios (5 seeds, 50 evaluation episodes, 95% Wilson upper bound 1.51%). The per-step safety-filter pipeline ($\sim$25 ms) is approximately ten times faster than the implemented SLSQP-based NMPC baseline (254 ms), and the filter intervenes proportionally to perturbation magnitude. The compositional $\epsilon(x)$ is empirically state-dependent under coupled perturbations and per-constraint differentiated by relative degree.

The method is intended as a commissioning-calibrated safety filter whose GP residual model should be trained on representative data before safety-critical deployment. The safety certificate applies to fixed, calibrated GP deployments within the commissioned operating envelope. This design should not be interpreted as a plug-and-play OOD-robust controller. Future work includes extending to matched uncertainty ($\Delta g \neq 0$), developing principled OOD detection for GP-based safety filters, and validation on higher-fidelity industrial simulators (SIL/HIL).

Statements and Declarations

Ethical considerations

Not applicable. This study did not involve human participants, human data, human tissue, or animal experiments.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Author contributions

Zhuang Shao: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Visualization, Writing – original draft, Writing – review & editing. Lijun Lei: Resources, Supervision, Project administration. Peng Wang: Formal analysis, Writing – review & editing. Liang Zheng: Resources, Supervision. Jie Zhou: Investigation, Data curation.

Data availability

The source code, simulation scripts, trained GP configurations, and plotting scripts required to reproduce the results are available at https://github.com/VickylastShao/Robust-HOCBF-Safety-Filtering-for-Supercritical-Power-Plants-under-Model-Mismatch.

Declaration of generative AI and AI-assisted technologies

During the preparation of this work, the authors used ChatGPT for language polishing and manuscript formatting. The authors reviewed and edited the content and take full responsibility for the published article.

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